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RESEARCH ARTICLE

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# Summary of final data from the 9-valent human papillomavirus vaccine pregnancy registry (2014–2021)

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**ABSTRACT**

The 9-valent human papillomavirus (9vHPV) vaccine targets human papillomavirus (HPV) types 6/11/16/18/31/33/45/52/58. Although its use in pregnant women is not recommended, inadvertent exposures during early pregnancy could occur as the indicated age range includes the ages of child-bearing potential. A pregnancy registry was created as a post-marketing regulatory commitment using spontaneous reports of 9vHPV vaccine exposure during pregnancy that were received from the United States (from December 2014 through March 2021) and followed through the completion of pregnancy. Primary pregnancy outcomes included congenital anomalies, live births, fetal deaths, elective terminations, and miscarriages. Enrolled prospective reports were used for rate calculations. Of the 180 prospectively enrolled pregnancies, 70 had known outcomes. There were 66 live births that resulted in 67 live born infants (95.7%; one twin pregnancy), one elective termination, and three miscarriages (estimated rate, 4.3% [3/69] of known outcomes excluding elective terminations). All three miscarriages occurred in the first trimester in women who had been vaccinated prior to detection of pregnancy. There were three major birth defects among the prospective reports, with an estimated rate of 4.5% (3/67). Findings from this analysis of the US 9vHPV vaccine pregnancy registry do not suggest an increased risk of major birth defects and miscarriage in women who received the vaccine. There is no apparent pattern of congenital anomalies to suggest potential adverse effects of exposure to the 9vHPV vaccine during pregnancy. This analysis is consistent with prior clinical trial experience and post-licensure observational data with the quadrivalent HPV and 9vHPV vaccines.

**PLAIN LANGUAGE SUMMARY**

The 9-valent human papillomavirus (9vHPV) vaccine can be given to prevent infection and diseases caused by HPV. Use of the 9vHPV vaccine during pregnancy is not recommended because there is limited safety information available. However, the recommended ages for women to receive the 9-valent HPV vaccine (9 to 45 years) includes those who can get pregnant, so there is a chance for women to be accidentally exposed to the vaccine during the early stages of pregnancy. As a result, it is important to collect additional safety information regarding vaccination during pregnancy. In the United States, a 9-valent HPV vaccine pregnancy registry was created to collect additional safety information on women vaccinated with the 9vHPV vaccine during pregnancy. Between December 2014 and March 2021, information on pregnancy outcomes was collected, including congenital anomalies, live births, fetal deaths, elective terminations, and miscarriages. Of the 180 pregnancies that were monitored, 70 known pregnancy outcomes were recorded, which comprised 67 live born infants from 66 live births (one pregnancy resulted in the birth of twins), one elective termination, and three miscarriages. All three miscarriages occurred during the first trimester of pregnancy in women who had been vaccinated before knowledge of their pregnancy. There was no apparent pattern of congenital anomalies to suggest potential adverse effects of exposure to the 9vHPV vaccine during pregnancy. These data do not suggest an increased risk of major birth defects or miscarriage in pregnant women who received the 9-valent HPV vaccine.

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## Introduction

The 9-valent human papillomavirus (9vHPV) vaccine (Gardasil 9; Merck Sharp & Dohme LLC, a subsidiary of Merck & Co., Inc., Rahway, NJ, USA) is a recombinant vaccine consisting of purified noninfectious virus-like particles of the HPV major capsid protein (L1) in an aluminum-containing adjuvant.[1–3] It targets HPV types 16/18/31/33/45/52/58, which cause ~90% of cervical cancers and HPV-related vulvar, vaginal, anal, and oropharyngeal cancers,[3,4] and HPV types 6/11, which cause ~90% of cases of genital warts and recurrent respiratory papillomatosis.[5,6] The 9vHPV vaccine was initially licensed in the United States in 2014[2] and is widely used and recommended.

Because the indicated age range of girls and women includes the ages of child-bearing potential,[2] inadvertent exposure to HPV vaccines during pregnancy can occur. Therefore, there is a need to monitor and assess outcomes of exposures to HPV vaccines during pregnancy. Results of animal toxicology studies have shown no evidence of teratogenicity, fetal harm, or effect on fertility associated with the 9vHPV vaccine[7] or quadrivalent vaccines for HPV (qHPV) types 6/11/16/18.[8,9] Available clinical trial and post-marketing data with 9vHPV[10,11] and qHPV vaccines[12–14] did not indicate increased rates of major birth defects or miscarriages in women inadvertently exposed to either vaccine during pregnancy. However, given the inherent limitations of such post hoc analyses, it is important to create prospectively designed pregnancy registries. One such registry conducted with qHPV vaccine[15] did not demonstrate adverse pregnancy outcomes related to the vaccine. This registry was a post-marketing commitment to the United States Food and Drug Administration which reported the final results of pregnancy data for the 9vHPV vaccine.[16] Enrollment into the United States Pregnancy Registry was completed in July 2022. This article presents the analysis of the final data for the pregnancy registry.

## Patients and methods

The pregnancy registry was established in the United States; however, spontaneous reports of exposure during pregnancy are received globally by the vaccine manufacturer and are entered into a global safety database for evaluation. Reports that originated in the United States were eligible for entry in the pregnancy registry. Patients were considered enrolled in the pregnancy registry if the following circumstances existed: (1) exposure was reported from the United States; (2) the patient was identifiable (presence of a unique patient identifier); (3) a healthcare provider (HCP) was identified; and (4) exposure to the 9vHPV vaccine occurred within 1 month before the date of onset of last menstrual period (LMP) or at any time during pregnancy. Reports that met these criteria were considered enrolled in the registry. A patient or provider could decline to participate in the registry at any time. As this was a post-marketing surveillance program, institutional review was not required. Pregnancy outcome information was obtained through completed questionnaires or telephone contact with the HCP. A report was considered lost to follow-up if no further information about the pregnancy outcome was available following three attempts to obtain information.

The primary outcomes of interest were congenital anomalies and pregnancy outcomes, consisting of live births, fetal deaths (defined as non-viable conceptuses in pregnancy ≥20 weeks after LMP), elective terminations, and miscarriages (defined as a spontaneous loss of a conceptus before the 20th week after LMP). To identify congenital anomalies not identified at birth, pediatric medical records were requested for all newborns for whom a signed maternal consent form was on file. Birth defects were defined according to the US Centers for Disease Control and Prevention Metropolitan Atlanta Congenital Defects Program (MACDP) guidelines.[17,18] Gestational age and birth weight were categorized according to definitions from the March of Dimes Foundation.[19,20] Full-term delivery was defined as 37–42 weeks from LMP; preterm births were defined as <37 weeks gestation; and very preterm births as those occurring prior to 32nd week completion. Low birth weight was defined as <2500 g (<5.5 lb) at term and very low birth weight as <1500 g (<3.3 lb). Timing of exposure was estimated from the first day of LMP and calculated for all doses falling within the defined exposure period (within 1 month before LMP or at any time during pregnancy). The first trimester was defined as 0–13 weeks from LMP, the second trimester was defined as 14–27 weeks from LMP, and the third trimester was defined as 28–42 weeks from LMP.

Reports were classified as prospective or retrospective. Prospective reports were those received before the pregnancy outcome was known, whereas retrospective reports were those received after the

outcome of the pregnancy was known. A report was classified as retrospective if an initial report was received after fetal testing identified an abnormality. Because of potential bias when reporting abnormal outcomes retrospectively, prospective reports were analyzed separately from retrospective reports. Annual registry reports, including cumulative data, were reviewed by a consultant teratologist to provide an independent evaluation of the possible etiology of birth defects, to assess for patterns, and to confirm the interpretation of case information for classification of major or minor congenital anomalies.

Enrolled prospective reports comprised the primary cohort for rate calculations. The estimated rate of congenital anomalies was calculated on prospective reports in the registry using the MACDP methodology: prevalence at birth, expressed as the number of reports with major defects per 100 live born infants.[17] Rates were not adjusted for maternal age. Reports of congenital anomalies were reviewed and evaluated by timing of exposure, biological plausibility, target organ, and expected population prevalence. Elective abortions were not included in rate calculations of all pregnancy outcomes because the outcome of a terminated pregnancy is unknown. The estimated miscarriage rate was determined by the number of miscarriages over the total number of known pregnancy outcomes, excluding elective terminations.

## Results

A total of 641 reports of maternal exposure to the 9vHPV vaccine during pregnancy were received from December 2014 through March 2021 and entered into the global safety database. Of these, 456 reports were not eligible for enrollment due to failure to meet one or more of the enrollment criteria: 339 were excluded due to the country of origin being outside the United States, and 117 US cases were excluded from registry enrollment (74 lacked patient identifiers or HCP information; 10 did not meet exposure criteria; HPV vaccine type was not identified in 16; registry participation was declined in 4; and 13 were ineligible for other reasons, including lack of reporter contact information). Overall, 185 (28.9%) reports met eligibility criteria for the pregnancy registry and were included in the analysis (Figure 1). Pregnancy outcomes for all those enrolled in the registry are presented in Table 1.

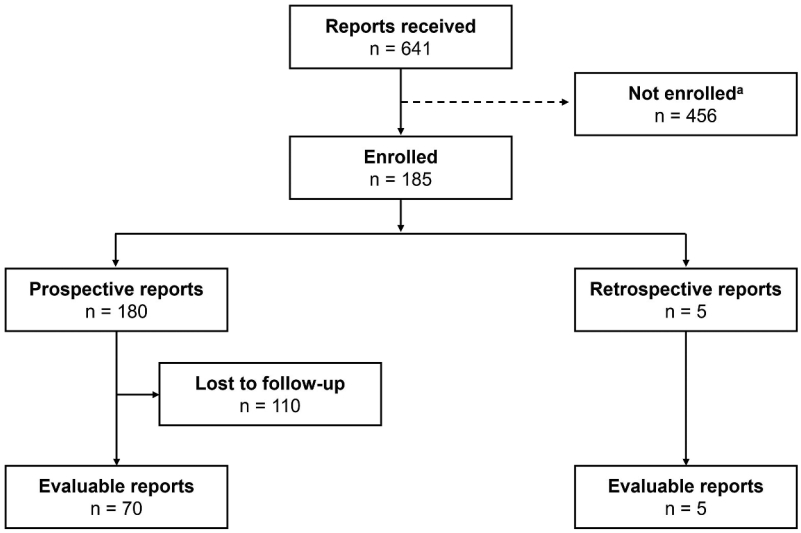



**Figure 1.** Summary of evaluable prospective and retrospective reports. [a]Did not meet enrollment criteria.

**Table 1.** Pregnancy outcomes in all enrolled reports (December 10, 2014 through March 31, 2021).

| Report type | Total reported pregnancies | Lost to follow-up[a] | Live births | Elective terminations | Miscarriages | Fetal death | Ectopic |
|---|---|---|---|---|---|---|---|
| Prospective | 180 | 110 | 66[b] | 1 | 3 | 0 | 0 |
| Retrospective | 5 | 0 | 5[c] | 0 | 0 | 0 | 0 |

[a]Pregnancy outcome unknown; [b]67 live born infants (includes one set of twins); [c]Six live born infants (includes one set of twins).

### Prospective reports

There were 180 enrolled and prospectively reported cases. Age of the vaccine recipient at the time of vaccination was reported in 152 cases and ranged from 12 to 38 years (median: 24 years). Age distribution was as follows: 34 women were 12–17 years; 84 were 18–29 years; and 34 were 30–38 years.

Of the 180 prospectively enrolled reports, 70 included pregnancy outcomes and 110 (61.1%) were lost to follow-up. A majority (n = 89) of cases lost to follow-up were due to unsuccessful follow-up with HCP or to no further information being provided by the HCP. Other reasons included transfer of the patient to another practice or location, an HCP report of unknown health status of the newborn, or inability of the HCP to locate the patient records.

As shown in Table 1, of 70 pregnancies with known outcomes, 66 exposed pregnancies resulted in live births of 67 newborns (95.7%; including one set of twins). Length of gestation was reported in 47 of these 66 pregnancies. Forty-six infants in 45 pregnancies (one set of twins) were born at term, one infant was born pre-term at 36 weeks' gestation, and one infant was born post-term at 43 weeks' gestation. Birth weight was reported for 43 of the 67 newborns. None of the birth weights met the definition of low birth weight (i.e., <2500 g).[19]

A single case of elective termination was reported and did not include a reason for the termination. Three miscarriages were reported. All three occurred in women vaccinated in the first trimester, prior to pregnancy detection, and no information was provided regarding examination of products of conception. The overall rate for miscarriage calculated from all pregnancy outcomes, excluding elective termination, was estimated at 4.3% (3/69). There were no reports of fetal death or ectopic pregnancy.

Timing of exposure to 9vHPV vaccine was reported in 65 of the 70 women with known pregnancy outcomes (Table 2). Most of those pregnancies (60/65) involved a single vaccine exposure; two were exposed within 30 days prior to LMP, 29 in the first trimester, 11 in the second trimester, and 18 in the third trimester. The remaining five women each noted two vaccine exposures during pregnancy.

Among the 67 infants resulting from the 66 prospectively enrolled pregnancies, 63 were born without major congenital anomalies, and three newborns had MACDP-defined major congenital anomalies, which are described in Table 3. One report noted an MACDP-defined major congenital anomaly of bilateral club feet that was diagnosed prior to *in utero* fetal exposure to 9vHPV vaccine. As the congenital anomaly was diagnosed prior to exposure to 9vHPV vaccine, this report is not included in Table 3 or the congenital anomaly rate calculation.

The three major congenital anomaly reports include one case each of congenital ureterocele, probable congenital hydronephrosis, and supraventricular tachycardia (SVT). Each of the three reports was reviewed by an independent teratologist who confirmed the MACDP classification. Using the 67 prospectively reported live born infants as the denominator according to MACDP methodology, the rate of congenital anomalies at birth was estimated as 4.5% (3/67).

Among the 63 infants born without MACDP-defined major congenital anomalies, one newborn had a nevus flammeus (port-wine stain), which is not included in the MACDP's classification as a major birth defect. Another newborn had abnormal genetic testing and fetal pyelectasis (transient prenatal hydronephrosis), but the condition was not confirmed after birth. These reports were not included in the calculation of the estimated rate of congenital anomalies.

**Table 2.** Timing of maternal exposure to 9vHPV vaccine, including multiple exposures, in prospective reports with known pregnancy outcomes.

| Outcome | Within 30 days prior to LMP | 1st trimester | 2nd trimester | 3rd trimester | Total no. of exposures (no. of outcomes) |
|---|---|---|---|---|---|
| Live births | 3[a] | 31[a,b,c] | 13[c,d] | 19[d] | 66 (61) |
| Miscarriages | 0 | 3 | 0 | 0 | 3 |
| Fetal death | 0 | 0 | 0 | 0 | 0 |
| Elective termination | 0 | 1 | 0 | 0 | 1 |
| Total no. of exposures | **3** | **35** | **13** | **19** | **70 (65)** |

[a]One pregnancy resulting in a live birth with exposure within 30 days prior to LMP had a second exposure during the first trimester; [b]Two pregnancies resulting in a live birth with both exposures during the first trimester; [c]One pregnancy resulting in a live birth with exposure in the first trimester had a second exposure in the second trimester; [d]One pregnancy resulting in a live birth with exposure in the second trimester had a second exposure during the third trimester.

Abbreviations: LMP, last menstrual period.

**Table 3.** MACDP-defined major congenital anomalies among infants of women[a] prospectively enrolled in the pregnancy registry.

| Congenital anomaly | Timing of exposure from LMP | Outcome | Notes |
|---|---|---|---|
| Congenital ureterocele | 2 weeks prior to LMP | Live birth | U/S at 36 weeks noted kidney-calyceal dilation. Infant born at term. Surgical procedure for congenital ureterocele after delivery.<br>Maternal medical history included 4 prior pregnancies, 1 live birth, and 2 elective terminations. Cervical dysplasia was identified as a maternal concurrent condition.<br>No genetic testing reported.<br>Maternal medication included betamethasone acetate (+) betamethasone sodium phosphate (celestone soluspan) and unspecified vitamins. |
| Hydronephrosis[b] | 5 weeks after LMP | Live birth | U/S at 34 weeks noted ureterovesical junction abnormality. Infant born at term and hydronephrosis was noted.<br>Maternal medical history included a motor vehicle accident and subsequent development of a brain aneurysm.<br>Quad and prenatal screening panel showed a Trisomy risk of 1 in 65.<br>Maternal vaccination with diphtheria toxoid, pertussis acellular vaccine, tetanus toxoid at 28 weeks after LMP and use of prenatal vitamins. |
| Supraventricular tachycardia (SVT) | 27 weeks after LMP | Live birth | Infant developed SVT at birth and multiple hospitalizations were reported. No other congenital anomalies reported.<br>Family history (grandparent) of SVT reported.<br>No genetic testing reported.<br>Maternal use of prenatal vitamins. |

[a]Each of these women was 20–29 years of age at time of pregnancy; [b]Report is categorized as probable hydronephrosis per MACDP guidance as no confirmatory postnatal U/S was reported.

Abbreviations: LMP, last menstrual period; MACDP, Metropolitan Atlanta Congenital Defects Program; SVT, supraventricular tachycardia; U/S, ultrasound.

### *Retrospective reports*

As shown in Table 1, there were five retrospective reports of 9vHPV vaccine exposure during pregnancy in the registry. In all five cases, outcomes of live birth (including one set of twins) were reported, and no major congenital anomalies were reported.

### *Non-enrolled reports*

Although the enrolled prospective reports of vaccine exposure during pregnancy comprise the primary cohort for rate calculations, all reports of exposure during pregnancy were reviewed for pregnancy outcomes and congenital anomalies.

Among US reports that did not meet enrollment criteria, one noted a live birth outcome with the reported congenital anomaly of gastroschisis. The LMP or estimated date of delivery was not reported; therefore, the timing of exposure to the vaccination could not be calculated with respect to pregnancy.

Of the 339 reports from outside the US, 331 were identified as confirmed exposure to the 9vHPV vaccine and there were 4 prospective reports describing abnormalities in the fetus/infant. Two of these four reports did not meet exposure criteria (within 1 month prior to the last menstrual period). Of the two remaining reports, one was from China and the other from Taiwan. While these two cases did not meet enrollment criteria for the pregnancy registry since they occurred outside of the US, they are described here for completeness.

The report from China described a pregnancy exposure at 2 weeks after LMP for which an induced abortion was performed at 33 weeks' gestation. The report noted that the umbilical cord was "deformed and spiral" and "the fetal heart rate was different." No further information was provided.

The report from Taiwan described a germ cell tumor in a 3-month-old male infant whose mother was given 9vHPV vaccine while pregnant. Timing of exposure was not provided, but based on the mother's vaccination date and the age of the infant at diagnosis, the exposure was estimated to be within the first 4 weeks after LMP. The infant underwent surgical intervention; no further information was reported. Insufficient information about the site of the neoplasm and tumor type precluded further categorization. This case was categorized under "neoplasms," which may be considered congenital anomalies when diagnosed prior to the sixth year of life, per MACDP guidelines.

## Discussion

In this pregnancy registry analysis, most pregnancies resulted in live births (66/70 prospective reports of maternal exposure), with no reports of fetal death. There were three reports of miscarriage (estimated rate, 4.3%) and one elective termination. Three infants were born with MACDP-defined major congenital anomalies (estimated rate, 4.5%); the other 63 infants, including one set of twins, were born without major congenital anomalies.

Given the inherent limitations and biases in this analysis, direct comparison to background rates in the general population is not advised. For example, it is expected that ascertainment of outcomes would differ between this analysis and other large population studies. By virtue of participating in a prospective cohort following vaccine exposure, it is more likely that birth outcome and a congenital anomaly, if present, would be captured and could result in a higher observed rate of anomalies than in the general population. In contrast, for the outcome of miscarriage, the nature of this surveillance (women must have remained pregnant up to the point of receiving vaccination) introduces a form of immortal time bias and may result in a lower observed rate for this outcome. Considering these limitations, background rates in the US general population are provided for context only; direct comparison was not performed. In the US general population, the estimated background risk of major birth defects and miscarriage in clinically recognized pregnancies is 2–4% and 15–20%, respectively.[2,21–23]

In this pregnancy registry, one case each of ureterocele and hydronephrosis was reported. Congenital anomalies of the kidney and urinary tract, including the aforementioned conditions, account for approximately 20–30% of all congenital anomalies identified in the neonatal period and in about 0.3–1.6 cases per 1000 live and stillbirths.[24,25] Fetal hydronephrosis, as detected on ultrasound, occurs in <4.5% of pregnancies.[26] Ureteroceles occur in approximately 1 in every 4000 children.[27] Single reports of ureterocele and hydronephrosis reported in the registry do not suggest a pattern of anomalies or a causal relationship to the 9vHPV vaccine.

Fetal SVT is a cardiac rhythm anomaly complicating approximately 1:1000 pregnancies.[28] Fetal tachyarrhythmias occur in approximately 0.4–0.6% of all fetuses;[29] of those, SVT is the most common, accounting for 70–80% of fetal tachycardia.[30] Fetal tachycardia and SVT are associated with congenital heart disease in approximately 1–5% of cases. SVT in infancy is thought to be due to an accessory atrioventricular connection that participates in a reciprocating tachycardia (reentry pathway), an intrinsic cardiac pathology.[31] All types of congenital heart disease have been found to be associated with SVT,[32,33] but in the case reported to the registry, no congenital heart disease or other anomaly was reported.

The results of this 9vHPV vaccine registry complement available data from clinical trials in which qHPV or 9vHPV vaccines[11,12] were administered prior to knowledge of pregnancy and post-marketing experience with both qHPV and 9vHPV vaccines.[10,13–15] The post-marketing data included a 6-year pregnancy registry experience with qHPV vaccine ( >1700 pregnancy outcomes), which did not show a higher rate of spontaneous abortions and major birth defects compared with the general population,[15] and cohort studies of pregnancies with 9vHPV or qHPV vaccine exposure, which did not identify an association of qHPV or 9vHPV vaccine exposure with spontaneous abortions or birth defects.[10,13,14] Consistent with these previous reports, results of this 9vHPV vaccine registry do not identify new safety concerns or suggest that inadvertent administration during pregnancy adversely affects pregnancy outcomes.

The findings in this registry are also consistent with recently published meta-analyses that show no association between inadvertent exposure to the 9vHPV vaccine during early stages of pregnancy and miscarriage.[34,35] A 2023 systematic review of seven articles (2009–2019), with three included in the meta-analysis (N = 1,380,424 individuals), showed that the rates of miscarriage and stillbirth after receiving any HPV vaccine during pregnancy were not significantly higher than in comparison populations.[34] HPV vaccination during pregnancy did not increase the risk of miscarriage (relative risk [RR] 2.01; 95% CI 0.66–6.13) or stillbirth (RR 2.02; 95% CI 0.65–6.27). However, lack of available studies and heterogeneity of the study populations limited analysis and interpretation of data.[34] In another meta-analysis of eight observational and four randomized controlled studies, no association was found between periconceptional or pregnancy exposure to HPV vaccination and miscarriage.[35]

This 9vHPV vaccine pregnancy registry analysis also has some limitations. Compared with the prior pregnancy registry for qHPV vaccine, the number of enrolled pregnancies is low. This may reflect several

external factors, including the current Advisory Committee on Immunization Practices recommendation to administer HPV vaccination at age 11 or 12 years, but can be given starting at age 9.[36] Thus, reporting may reflect vaccine use in age groups in which pregnancy exposures are unlikely. In addition, one of the key limitations for this registry is the loss to follow up for most of the prospectively enrolled reports despite diligent efforts for follow-up (3 phone call attempts). When pregnancy outcomes were reported, to identify defects that may not have been evident at birth, newborn pediatric medical records were requested; however, the pediatric medical records were generally not provided. Additionally, the data collection period for the registry encompasses the COVID-19 pandemic (2020–2022), which may have impacted reporting and follow-up. Although data from the registry are reassuring, the generalizability of the US experience to other countries may be limited.

Due to the nature of voluntary spontaneous reporting, no definitive conclusions can be made based on the 9vHPV vaccine pregnancy registry data alone. The totality of the available data should be considered in the assessment of the safety of the 9vHPV vaccine during pregnancy. Although available data are insufficient to support the recommendation of vaccination during pregnancy, they do not suggest that inadvertent exposure to the 9vHPV vaccine adversely affects pregnancy outcomes, such as major birth defects or miscarriages.

In conclusion, the results of this US pregnancy registry analysis (post-marketing experience from 2014–2021) for the 9vHPV vaccine do not suggest an increased risk of major birth defects and miscarriage in women who receive the 9vHPV vaccine. These results are consistent with prior results from clinical trials of 9vHPV and qHPV vaccines, as well as post-licensure observational data with the qHPV vaccine.[10–15] The totality of data from clinical trials and post-marketing experience (including pregnancy registries) does not suggest that 9vHPV vaccine administration adversely affects pregnancy outcomes (major birth defects and miscarriages). Case reports of exposure to the vaccine during pregnancy continue to be closely monitored as part of routine pharmacovigilance.

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## Disclosure statement

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests.

AM, BS, DR, BS, RL, JET, AL, and SK are employees of Merck Sharp & Dohme LLC, a subsidiary of Merck & Co., Inc., Rahway, NJ, USA, and may own stock and/or stock options in Merck & Co., Inc., Rahway, NJ, USA. CH was an employee of Merck Sharp & Dohme LLC, a subsidiary of Merck & Co., Inc., Rahway, NJ, USA, at the time of this registry, and owns stock in Merck & Co., Inc., Rahway, NJ, USA.

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## Data availability statement

The data sharing policy, including restrictions, of Merck Sharp & Dohme LLC, a subsidiary of Merck & Co., Inc., Rahway, NJ, USA (MSD), is available at https://trialstransparency.msdclinicaltrials.com/policies-perspectives.aspx. Requests for access to the clinical study data can be submitted via e-mail to the Data Access mailbox.

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